

it's game over for you. This is because the latest software update that Apple has rolled out, chooses convenience for the user, over security. The thief can change the password for the Apple ID, access the iCloud data and tackle the two factor authentication easily as well. The thief can then remotely wipe the data of all your other Apple devices as well. Details of banking accounts, PINs, passwords to gaming apps, details for email accounts are all different kinds of sensitive information that you are better off not feeding into your smartphone in a public area.

Say a thief has stolen your device, but does not know the PIN or unlocking pattern, there is still a method to access the device and disable the security measures, such as GPS tracking or remote wiping of data. Researchers from the University of Pennsylvania have demonstrated a technique which uses one of the fundamental side effects of touchscreen use. Human

fingers have oily skin, and touchscreen use leaves behind smudges. The technique is known as a smudge attack, and involves deciphering the password sequence by photographing the smudges on a device in the ideal lighting conditions. The smudges not only show which are the touchpoints of the password pattern, but also the swipe patterns. The attack method works even after using other apps on the same screen, and even after the screen has come into contact with clothing. The passwords are used so frequently, that the underlying patterns remain on the screen. The pattern passwords on Android devices are particularly susceptible to this kind of attack. The team is going on to study human approaches to pattern selection, as well as study the heat trails left on the screens by the movement of fingers, to further improve their methods.

While some of the newer devices have done away with fingerprint sen-

sors, a number of phones still use the finger to authenticate access to a device. All an attacker needs to access the device is a high resolution photo of your finger, or a fingerprint lifted from one of the objects you have touched. This data can then be used to 3D print a high resolution version of your finger, which can be used to access the device. A team of researchers from the Michigan State University have demonstrated the technique on all kinds of fingerprint sensors on phones - optical, capacitive and ultrasonic.

Facial recognition is another new biometric authentication feature in smartphones. OnePlus does not encourage using the facial recognition for sensitive applications, such as online payments. The FaceID on the Apple iPhone X has been fooled with a 3D printed mask of the person. The most sophisticated technologies in the world are susceptible to the extremely mundane attacks. ■

MALEFICENT MALWARE

While a lot of malware has been discussed elsewhere in this article, there's still a significant need to understand it deeply and what it is capable of beyond the examples. Especially since it is very easy for Malware to reach us, such as over an app installed on our phone.

One of the latest and rising techniques in the world of malware is fileless malware. This type of malware is essentially invisible to most detection tools. Fileless malware infections use the RAM and the Kernel and do not require any file to be downloaded onto the user hard drive. The entire execution is done within another execution. Persistent threats in the BIOS or UEFI can remain on the machine without being detected by antivirus software, and re-infect a system even after it has been scrubbed. One of the easiest opportunities for threat actors is to run code through the browser window, that can compromise both desktops and smartphones at the same time.

Attackers have also realised that sensitive and lucrative targets have realised the risks and upped their security levels to unfeasible standards. Persistent attackers will be looking into the supply chain for that system and once they find a third party with vulner-

abilities, they'll be able to target it to gain access to the system itself. Malware target identification and delivery will both be automated to a large extent this year using AI and machine learning. On the other hand, it will also be available as a service more easily, thanks to the dark web. Ready to use exploit kits will also be available in greater numbers this year.

An interesting way to avoid malware infections is via 'vaccination'. Many malware strains, like the ransomware Wannacry, left identifying markers on target machines where they'd executed once, so as not to target it again. Security researchers have recommended the identification of such markers and using them to 'vaccinate' your machines. However, this approach does not hold too much promise considering that even the basic patches are not applied to the machines.

There can potentially be new ways in which cybercriminals will use extortion, beyond the use of ransomware, and we could see the emergence of cybercriminals using blackmail, disinformation campaigns, or threaten to release highly sensitive, or compromising data to the public. Combined with blockchain technologies, cloud computing and artificial intelligence, cybercriminals will have an increasingly sophisticated arsenal at their disposal. ■